COMPANY CASE STUDY

IMAGE PROCESSING AT AMAZON

Visual Search, Product Integrity, Intelligent Logistics and Checkout-Free Retail

Submitted by	Apurvasingh Chauhan
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EXECUTIVE SUMMARY

Image processing has become an operational capability in digital commerce: it can help a customer identify a product from a photograph, help a marketplace screen images and labels for risk, inspect goods as they move through fulfilment, and support checkout-free retail. This report examines Amazon because the course material itself uses Amazon as an e-commerce image-processing case study, and because the company has publicly described several relevant systems.

The report analyses five connected applications: Amazon Lens and Lens Live visual search; Project PI product inspection; image-based intellectual-property and safety-label checks; computer-vision-assisted package handling; and Just Walk Out checkout-free shopping. Across these cases, the common pattern is not a single algorithm but a workflow in which image capture, preprocessing, computer vision, contextual data and an operational decision are combined. The most important business value appears when image analysis changes a real process - for example, surfacing a product match, holding a potentially damaged item, or guiding a worker to the correct package.

The case is balanced. Amazon reports substantial scale, including billions of images inspected by Project PI in 2025 and Just Walk Out operating at more than 360 third-party locations across five countries as of May 2026. These are company-reported figures, not independent performance audits. Amazon also announced the closure of its Amazon Go and Amazon Fresh physical stores in January 2026, while continuing to expand Just Walk Out as a technology service. This illustrates an important lesson: technical capability and commercial success of a particular retail format are not the same thing.

The report concludes that Amazon's strength lies in embedding visual intelligence into retail and logistics workflows, while its continuing challenges include accuracy under occlusion and variable lighting, false decisions, privacy, transparency, human oversight and measurable unit economics.

1. Introduction and company context

Digital image processing converts visual input into data that can be enhanced, compared, classified or used to support a decision. In a commercial setting, the objective is rarely to "process an image" for its own sake. The objective may be to improve product discovery, detect a defect, verify a label, identify a parcel or determine which product has been removed from a shelf. Image processing therefore sits inside a wider business workflow that includes cameras, devices, catalogues, sensors, people, software services and operational rules.

Amazon is a suitable case because its retail and operations ecosystem contains large volumes of product images, customer-submitted visual searches, fulfilment-centre imaging and physical-store camera systems. The company's 2025 Form 10-K reports consolidated net sales of approximately US$716.9 billion for the year ended 31 December 2025: US$426.3 billion in North America, US$161.9 billion internationally and US$128.7 billion in AWS. These are company-wide segment figures, not revenue attributable to image processing; they show the operating scale in which visual technologies are embedded (Amazon.com, Inc., 2026).

The course lecture identifies multiple Amazon business units where image transformations and computer vision can be relevant, including Seller/Catalog, Amazon Robotics, Amazon Go, Prime Air, brand protection, Amazon Photos, Advertising and Ring. It also highlights the importance of sequencing rotation, cropping, resizing and translation for a specific business task (Patel, n.d.). This report uses that teaching framework, but does not treat illustrative classroom values or hypothetical transformation sequences as Amazon's actual production specifications.

1.1 Scope and research method

The analysis uses publicly available Amazon and AWS descriptions, Amazon's 2025 Trustworthy Shopping Experience Report (published April 2026), the company's 2025 annual report, and independent reporting on the January 2026 grocery-store decision. The strongest evidence is where Amazon describes a named system and its function. Where the company does not publish model architecture, operating thresholds, error rates or image-processing parameters, the report does not invent them. Instead, it separates documented capability from technical interpretation and recommendations.

A key limitation is source bias: many technical descriptions and headline metrics are published by Amazon itself. They are useful primary sources for what the company says it operates, but they do not independently establish accuracy, customer outcomes or return on investment. Accordingly, all scale and pilot-performance figures are labelled as Amazon-reported where applicable.

1.2 A common image-processing workflow

Across the cases, the conceptual pipeline is: capture an image; check and prepare it; detect or recognize objects, text or visual features; combine the visual result with contextual information; and trigger an action. Typical preparation operations taught in the course include rotation, cropping, resizing and translation. Colour-space conversion, grayscale analysis, edge detection, texture features and image-quality metrics may also be useful in computer-vision systems. However, Amazon does not publish enough implementation detail to claim that every named technique is used in every product. Figure A1 in Appendix A summarizes this common workflow as an analytical model, not a diagram of Amazon's confidential architecture.

2. Visual product discovery: Amazon Lens and Lens Live

Traditional e-commerce search begins with words: a customer names a product, describes its features or enters a brand. Visual search changes the starting point. A customer can photograph an unfamiliar object, upload an existing picture or scan a barcode, and use the image as a query. Amazon Lens on Amazon.in supports photo upload, camera search and barcode scanning, returning similar products that a customer can compare (Amazon Staff, n.d.-a). This is particularly useful when a shopper knows what an item looks like but does not know its product name, model or correct search terms.

Amazon's September 2025 announcement of Lens Live describes a more immediate experience. The feature scans products through the smartphone camera and displays matching products in a swipeable carousel. Customers can tap an object in the camera view, add a match to the cart or save it to a wish list without leaving the camera interface. Lens Live also integrates Amazon's AI shopping assistant for summaries and product questions; when an exact or similar product is not identified, the system can display an intelligent caption describing the visible scene (Doerr, 2025).

2.1 What image processing contributes

Amazon states that Lens Live uses a lightweight object-detection model running on the device to identify products in real time. It then uses a deep-learning visual-embedding model to match the camera view against Amazon's product catalogue, which Amazon describes as containing billions of products. The service is deployed using AWS-managed OpenSearch and Amazon SageMaker services (Doerr, 2025). In simple terms, the detection stage helps locate the likely object in the camera frame, while visual matching retrieves catalogue items that look similar. The final result is a ranked product-discovery experience, rather than a conventional text-results page.

This illustrates the connection between image representation and search. A visual model can encode an image or object crop into a numerical representation that supports similarity matching. The actual model design, embedding dimensions, similarity thresholds and ranking formula are not disclosed in the public announcement, so they should not be assumed. The important published fact is that the product combines on-device object detection with visual-embedding search and shopping-assistant context.

2.2 Business value and limitations

The expected business value is lower search friction: customers can move from "I can see it" to "I can find a similar product" with fewer steps. Visual search can also help customers discover substitutes when the photographed object is not sold in the exact form shown. Its quality depends on the image: blur, poor lighting, clutter, partial occlusion, unusual viewpoints, reflections and visually similar products can reduce match quality. Product listings can also be incomplete or contain inconsistent images. A wrong match is not merely a computer-vision error; it may lead to an irrelevant purchase or a return.

For India, Amazon Lens is directly relevant because Amazon.in supports photo, camera and barcode-based search. Lens Live, however, was announced for U.S. customers using the Amazon Shopping app on iOS and Android; this report does not claim that Lens Live is available in India (Amazon Staff, n.d.-a; Doerr, 2025).

3. Product integrity: Project PI and automated inspection

A product can be correctly ordered and still disappoint the customer if it is damaged, expired or otherwise unsuitable when it reaches the packing and dispatch stage. Amazon's Project PI (Private Investigator) addresses part of this problem by using imaging tunnels already present in fulfilment centres. These tunnels were designed to capture images as products move through the network; Project PI adds a quality-inspection purpose to the existing image stream (Amazon, 2026a).

Amazon describes two direct visual checks. First, computer vision examines product images to help confirm that expiration dates have not passed. Second, it compares the actual product image with a reference image from the product catalogue to identify visible differences such as a bent book cover or a cracked screen. Amazon also says generative AI analyses the product images together with customer feedback to support corrective action. When a potential issue is identified, affected units can be removed from inventory before shipment (Amazon, 2026a).

3.1 Why image comparison is useful

The catalogue image provides a reference for what a product should look like. The captured image is the observation of the physical item at a particular point in the fulfilment process. A comparison system can look for differences in shape, surface appearance, packaging condition or visible markings. This is conceptually related to image registration and image-quality comparison: two images must be sufficiently aligned and comparable before differences are meaningful. If viewpoint, scale, lighting or packaging version varies, the comparison becomes harder. Amazon has not published the exact registration method, defect classifier, confidence threshold or image-quality metrics used by Project PI.

The course covers MSE, PSNR and SSIM as image-comparison metrics. These illustrate three ways to compare pixel values or visual structure, but this report does not claim that Amazon uses those exact metrics in Project PI. In a proposed evaluation design, MSE could identify large pixel-level differences, while SSIM could help assess structural similarity. Neither metric alone is a complete defect detector: a small but safety-critical crack may occupy few pixels, while harmless lighting differences may produce a large pixel error. Defect-specific models and human escalation remain important.

3.2 Reported scale and operational impact

In its April 2026 Trustworthy Shopping Experience Report, Amazon states that its automated system inspected billions of images in 2025. This is a significant scale claim, but it is company-reported and is not accompanied in the report by independently audited precision, recall, false-rejection rates or cost savings. Amazon's wider product-verification work also includes metrology imaging and X-ray inspection for issues that cannot be seen on the outside of a package. The company itself distinguishes visible defects that cameras can catch from hidden characteristics requiring deeper inspection (Amazon, 2026a).

The business logic is preventive: identify a likely defect while the product is still inside the network, rather than allowing it to create a customer complaint, refund, replacement or trust problem after delivery. The limitation is equally important: ordinary optical imaging cannot reveal every internal defect, and a product that looks visually correct is not necessarily safe or genuine.

4. Image-based product safety and intellectual-property protection

Amazon's 2025 Trustworthy Shopping Experience Report describes visual analysis as one component of its wider marketplace-protection controls. The company says its automated systems scan billions of attempted product-detail-page changes daily for signs of potential abuse. It specifically reports scanning text, logos and images to detect possible use of registered trademarks or copyrighted works in infringing ways. Amazon also says that multimodal models can analyse visual elements together with text and pricing patterns, helping identify cases in which logos or images have been manipulated or obscured (Amazon, 2026a).

The report states that Amazon's proactive controls blocked more than 99.9% of suspected infringing listings before a brand owner had to find and report them. This figure is an Amazon-reported outcome; the public report does not provide a full independent evaluation method, denominator definition or false-positive rate. It should therefore be understood as the company's stated performance rather than as a verified benchmark that can be compared directly with another platform (Amazon, 2026a).

4.1 Omniscan: six-sided product imaging

Amazon also describes Omniscan, a machine-learning system that captures and analyses six-sided images of products inside fulfilment centres. Its stated purpose is to check whether important safety information is readable and in the appropriate language, including brand names, addresses and product warnings. Amazon says Omniscan units have been deployed since 2024 across parts of its fulfilment network in the United States, Canada, the United Kingdom, Turkey, Saudi Arabia and Europe, producing image sets for more than 12 million products (Amazon, 2026a).

This is a practical use of controlled image capture. Multiple views increase the chance that a label on a side or rear panel is visible. Computer vision can then locate text regions and assess readability; optical character recognition can support text extraction, while language and compliance rules can check whether required information is present. The company's public description confirms image capture and AI verification, but it does not disclose the OCR engine, confidence scores or exact compliance rules. These implementation details should not be invented.

4.2 Why visual evidence is not enough

A counterfeit product can imitate a logo or package design, while a genuine product can have a damaged, replaced or outdated label. Image analysis is therefore strongest when it is combined with other signals: seller identity, catalogue history, customer reports, serialized product identifiers, supply-chain records and specialist examination. Amazon's report describes such a layered approach, including Transparency serialization and laboratory analysis. It notes that more than 2.7 billion product units have been verified as genuine through its Transparency programme (Amazon, 2026a).

For business operations, the main advantage of image-based screening is scale and early detection. The main risk is treating a visual match as proof of authenticity or compliance. A robust process should distinguish "potentially suspicious" from "confirmed infringement," route uncertain cases to trained reviewers, and provide a correction or appeal path for legitimate sellers.

5. Intelligent logistics: computer vision-assisted package handling

Image processing also supports the movement of goods after they leave the fulfilment centre. Amazon has publicly described Vision-Assisted Package Retrieval (VAPR), a system in delivery vans that locates and decodes multiple package barcodes in real time. Machine-learning models are trained to recognize different labels and packages under changing lighting conditions and package characteristics. The system uses visual cues, including a green symbol for packages to deliver at the stop and red shapes for other packages, helping the driver identify the right parcels without repeatedly checking a handheld device (Amazon Staff, 2025a).

Amazon reports that early VAPR tests reduced drivers' perceived physical and mental effort by 67% and saved more than 30 minutes per route. These are results reported by Amazon from early testing, not an independent controlled trial. Even so, the example shows how a visual model can be connected to a simple physical interface: the camera interprets labels and package context, while a projection directs a human operator's attention to the relevant parcel (Amazon Staff, 2025a).

5.1 Vision Assisted Sort Station (VASS)

VASS extends the same idea to delivery stations. Amazon describes a camera-based system that scans packages as they move on a conveyor and projects visual cues to help employees match a package with the appropriate route bag. Destination bags can be brought to the worker, reducing the need to walk between locations or repeatedly look at a screen. VASS is explicitly described as being in a pilot phase, with implementation across some delivery stations in Europe and the United States scheduled to start in 2027; it should not be presented as universally deployed (Amazon Staff, 2025a).

During early testing at Amazon's Innovation Center in Dortmund, the company reported that workers could manage a buffer of up to ten packages simultaneously, compared with two under conventional methods. Amazon also reported lower mental load and fewer missorts in pilot stations. These are pilot observations, not final network-wide statistics. The system's business value is not only faster sorting: it may also reduce cognitive burden and make the task easier to perform consistently (Amazon Staff, 2025a).

5.2 Link to image-processing concepts

Package recognition illustrates object detection, barcode or text reading, image localization and decision support. Geometric transformations such as cropping and resizing can be useful in preparing label regions for recognition, while image-quality checks can flag blur, glare or insufficient resolution. The exact VAPR/VASS preprocessing sequence is not public. The course worksheet's transformation examples are useful for understanding why orientation, cropping and resizing order can affect accuracy, but its example angles and image dimensions are teaching scenarios rather than published Amazon production settings (Patel, n.d.).

A practical limitation is that labels can be folded, covered, damaged, duplicated or poorly illuminated. A system that makes an incorrect parcel-to-route assignment can create delivery delay or a misdelivery. Human confirmation, visible exception handling and monitoring of missort rates are therefore essential parts of the system design.

6. Checkout-free retail: Just Walk Out and Dash Cart

Just Walk Out is one of Amazon's best-known computer-vision applications. A shopper first links a payment method to a shopping trip, then picks up or returns products from shelves. Cameras and computer-vision models track interactions with merchandise and update a virtual cart. Object-recognition models help identify the products. Shelf weight sensors provide additional evidence for small or hard-to-see items, and sensor fusion combines these signals to determine which items were selected (Amazon Staff, n.d.-b).

Amazon has described a newer multimodal foundation model that analyses camera and sensor inputs simultaneously rather than processing a customer's movements and product interactions only as a sequential chain. The stated aim is to handle difficult conditions such as camera obstruction, poor lighting, nearby shoppers, items being picked up and returned, or product rearrangement. Amazon says the system uses a 3D map of the store and an image catalogue of merchandise to understand the layout and products (Jenkins, n.d.).

Amazon states that Just Walk Out does not collect biometric information for the shopping-identification process. It links a payment method to a temporary trip identifier, tracks where shoppers and their hands interact with shelves, and discards the temporary code after the visit. The company also says associates label and annotate real and synthetic data to improve the models, but do not watch live video to generate receipts (Amazon Staff, n.d.-b; Amazon, 2026b). These are Amazon's descriptions of its own system and privacy practices.

6.1 Commercial scale and strategic reset in 2026

The latest strategic context matters. On 27 January 2026, Amazon announced that it would close its Amazon Go and Amazon Fresh physical stores and convert various locations into Whole Foods Market stores. Amazon said its branded physical grocery formats had not yet produced the distinctive customer experience and economic model needed for large-scale expansion. Independent reporting by Reuters also described the decision as a shift in Amazon's grocery strategy (Amazon, 2026b; Reuters, 2026).

This does not mean that the computer-vision technology was abandoned. In an update dated 12 May 2026, Amazon reported that Just Walk Out continued to operate in more than 360 third-party locations across five countries, and that more than 40 North American fulfilment centres were using it in breakrooms. The distinction is important: Amazon stepped back from operating its own Go and Fresh physical-store formats while continuing to commercialize checkout-free technology for other venues and internal use (Amazon, 2026b).

The lesson is that a technically impressive experience does not guarantee that a particular retail format has attractive economics. Hardware installation, model maintenance, store layouts, customer habits, error resolution and the value of saved checkout time all affect viability. Dash Cart, which adds computer-vision-based item tracking to a smart cart, represents a different format for larger grocery shops; it should not be confused with the camera-and-sensor infrastructure used in every Just Walk Out location (Amazon, 2026c).

7. Cross-case analysis, risks and recommendations

The five applications show how image processing supports several stages of commerce. Lens assists discovery; Project PI inspects products; intellectual-property and Omniscan checks support marketplace integrity; VAPR/VASS connect image understanding to package movement; and Just Walk Out turns camera and sensor observations into a transaction. The common value chain is capture, image preparation, visual interpretation, context fusion and an operational action. Table A1 in Appendix A compares these use cases.

7.1 Business benefits

The main benefits are: (1) reduced friction in product search; (2) earlier identification of visible product defects or potentially non-compliant labels; (3) proactive detection of suspicious listings; (4) more efficient parcel sorting and retrieval; and (5) faster store transactions. Importantly, the value comes from embedding the model in an end-to-end workflow. A detection result that does not reach a catalogue action, inventory hold, worker cue or receipt process has limited business value.

7.2 Risks and governance considerations

Accuracy is affected by lighting, blur, occlusion, reflections, camera position, product changes and crowded scenes. False negatives can allow a damaged item, wrong parcel or incorrect charge to pass through; false positives can block legitimate products, create unnecessary inspections or delay delivery. Marketplace protection creates an additional fairness concern: an automated system must distinguish a genuine infringement from a lawful resale, a permitted use or a visually similar but legitimate product. Human review and an appeal path are necessary for high-impact decisions.

Privacy and transparency are particularly important for in-store camera systems. Even when a system does not use facial recognition, it can still observe movements and interactions in a physical space. Customers should receive clear notice, data should be limited to what is needed, access and retention should be controlled, and incident response should be defined. Employee-facing systems should be designed with worker consultation, ergonomic evaluation and training rather than treating labour impact as a secondary matter.

7.3 Recommendations

- Create an image-quality gate at capture: check blur, exposure, glare, occlusion, orientation and minimum resolution before an image enters a high-impact workflow.
- Use human-in-the-loop exception queues for uncertain defects, suspected infringement, safety-label failures and ambiguous retail interactions; track overrides and reviewer consistency.
- Measure performance by task and context: precision, recall, false-positive and false-negative rates, latency, missort rate, inspection coverage, incorrect-charge rate and customer correction rate. Report results by product category, store layout and lighting condition.
- Use reference-image comparisons carefully. MSE and SSIM can help monitor image similarity, but should be combined with defect-specific validation and operational outcomes; a single pixel metric should not decide safety or authenticity.
- Publish clear privacy notices and governance controls for camera-based systems, and periodically audit data access, retention, employee impact and customer dispute processes.

8. Conclusion

Amazon demonstrates that image processing is not limited to editing photographs. It is a business capability that connects visual evidence with search, product quality, compliance, logistics and retail transactions. Its strongest pattern is the combination of computer vision with catalogue, sensor, customer and operational data. At the same time, Amazon's 2026 grocery-store reset shows why technology outcomes must be evaluated alongside customer preference and unit economics. For business leaders, the central lesson is to design visual systems around a clearly defined decision, validate them under real operating conditions, retain human oversight where errors matter, and measure the business outcome rather than the novelty of the model.

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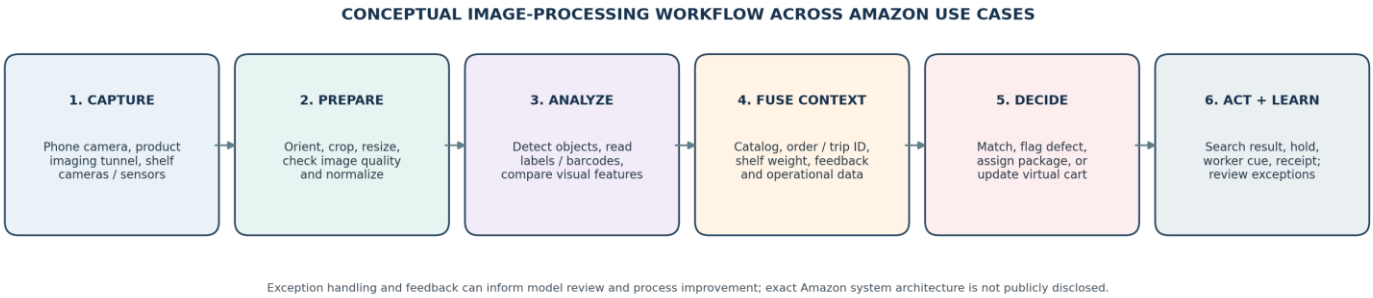
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Appendix A. Visual workflow and comparison tables

Figure A1. Conceptual image-processing workflow across Amazon use cases



Note. *This is an analytical synthesis of public system descriptions, not Amazon's disclosed internal architecture.*

Table A1. Summary of documented image-processing applications

System	Visual input	Processing task	Operational use
Amazon Lens / Lens Live	Phone photo or live camera view	On-device object detection; visual-embedding match to product catalogue	Product matches, comparison and shopping actions
Project PI	Images from fulfilment-centre imaging tunnels	Visual inspection; comparison with catalogue references; image + feedback analysis	Potentially damaged / expired units flagged and removed from inventory
IP checks / Omniscan	Product-page images and six-sided product images	Scan logos/images/text; assess safety-label readability and language	Potential infringement or product-compliance issue surfaced for action
VAPR / VASS	Package and label images in vans / delivery stations	Barcode and package recognition; visual cues / projections	Driver or worker guided to the correct parcel / route bag
Just Walk Out	Store cameras plus shelf sensor data	Object recognition and multimodal sensor fusion	Trip-linked virtual cart and automated checkout

Source: Amazon public product and technology descriptions (Amazon, 2026a, 2026b; Amazon Staff, 2025a, n.d.-a, n.d.-b; Doerr, 2025; Jenkins, n.d.).

Table A2. Suggested performance and governance controls

Control area	Example risk	Recommended measures
Visual quality	Blur, glare, occlusion, poor orientation	Capture-quality pass rate; reject / recapture rate; performance by lighting and view
Model errors	Missed defects, wrong product match, incorrect parcel assignment	Precision, recall, false-positive / false-negative rate; human override and appeal rate
Operational reliability	Latency, scanner downtime, camera obstruction	P95 latency; uptime; missort / incorrect-charge / delayed-inspection rate
Privacy and trust	Excessive observation, unclear retention or purpose	Data minimization; retention audit; access-log review; customer notice and dispute process
Workforce impact	Increased monitoring or poorly designed worker prompts	Worker feedback; ergonomic assessment; training completion; task-time and error trends

Interpretation: these are recommended evaluation measures for a responsible image-processing programme; they are not presented as Amazon's published internal KPI set.